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Palaeomagnetic constraints on formation of the Mianwali reentrant, Trans-Indus and western Salt Range, Pakistan

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Successions of Lower to lower Middle Cambrian, Upper Permian to Upper Triassic and Lower Tertiary carbonates and arenites have been sampled in five sections, representative of the three main segments of the Mianwali reentrant in the (Trans-Indus) Salt Range (northern Pakistan), i e the southern Khisor Range, the northern Surghar Range and the western Salt Range Comparison of primary and secondary magnetization directions with the Indian APWP demonstrates the secondary origin of the Mianwali reentrant and shows a pattern of rotations which varies in sense and magnitude along the reentrant with the main structural trends Data from the Trans-Indus and western Salt Range and published Early Cambrian, Early Permian and Late Tertiary palaeomagnetic results from the southern Salt Range and the Potwar Plateau show that the Hazara Arc underwent a 20–45° counterclockwise rotation relative to the Indian Shield A contrasting clockwise rotation over about 45° has recently been established for thrust sheets in the opposing eastern limb of the Western Himalayan Syntaxis, i e for the Panjal Nappe [1] and the Riasi thrust sheet [2] These palaeomagnetically established rotations conform with the about 75° azimuthal change in structural trend along the Syntaxis, and support Crawford's [3] suggestion that the Salt Range was originally in line with the northwestern Himalaya The Salt Range front prograded and moved southwards as part of the Hazara Arc thrust sheet, detached from basement along the evaporitic Salt Range Formation The Mianwali reentrant originated through obstruction of the southwards advancing thrust sheet by moulding around basement topography of the northwest oriented Sarghoda Ridge

1. Introduction

The north-south trending Pakistan Fold Belt (PFB; Figs 1, 9) forms a remarkable interruption of the latitudinally oriented Alpine-Himalayan belt Together with the Himalayan Arc, the PFB and its mirror image in the Arakan Yoma of northeastern India and Burma reflect large-scale northward impingement of Greater India into south-central Asia. This occurred from about the Cretaceous/Tertiary time boundary onwards, with the suture zone moving from equatorial latitudes to about 30° N [2,4,5]. The main structural trends of the collision belts, however, are of Late Tertiary origin The Himalayan Arc is mainly characterized by post-Oligocene intracontinental underthrusting along the Main Central Thrust (MCT), with oroclinal bending since latest Miocene [4] The present-day surface outline of the PFB (Figs 1, 9) with its fold festoons and intervening syntaxes results from Plio-Pleistocene foreland movements.

There are pronounced structural differences between the PFB and the Himalaya. The Himalaya is characterized by doubling of the crust in the Tethyan Himalayan region, metamorphism and thrust-stacking in the Lower Himalaya, and detachment of the sedimentary cover in the foreland The PFB, in contrast, does not show significant metamorphism or intracontinental underthrusting Its fold festoons and reentrants (Fig. 9) are of a more local and varied tectonic origin.

Various authors [6–8] have argued that the arcuate structure of the Sibi and the Bannu reentrant (Figs. 1, 9), which incorporates the Mianwali reentrant, resulted from deflection of the foreland

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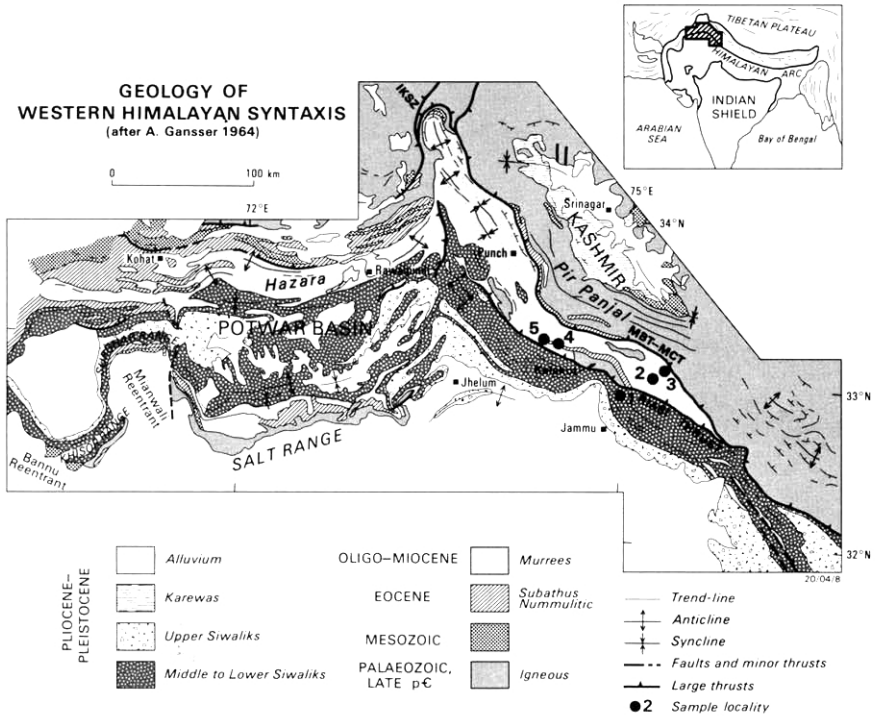


Fig 1 Geological map of the Western Himalayan Syntaxis after Gansser [40] Sampling localities are indicated by full dots See Table 1 *1* = Saidu Valley, *2* = Lunda Gorge, *3* = Chichali Gorge, *4* = Nammal Gorge, *5* = Dhak Pass Inset PFB = Pakistan Fold Belt, AY = Arakan Yoma

thrust belt through basement ridges and young molasse wedges [9] Their mode of origin may be significantly different from the Western Himalayan Syntaxis (Fig. 9). Wadia [10] attributed the extremely tight U-form in plan of the latter to a basement promontory dividing the Himalayan front into two components with moulding of structures It is more plausible, though, that the presence of the Infra-Cambrian evaporitic Salt Range Formation beneath the Hazara Arc and absence or less extensive development of evaporites in the Himalayan region [6,11,12] facilitated an exceptionally wide detachment zone in the Hazara Arc region and induced the U-form of the Western Himalayan Syntaxis (Fig. 9) [12,13]

Whilst a secondary origin for the festoons and reentrants of the PFB is generally assumed on structural grounds, supporting quantitative and qualitative palaeomagnetic evidence has been produced so far only for parts of the Sulaiman lobe [8] and the Salt Range–Potwar Plateau region [14–18] A counterclockwise rotation of the southern Salt Range over 20–30° relative to the Indian Shield was suggested [19] from comparison of Cambrian and Permian palaeomagnetic data with the general Gondwana APWP (Fig. 7). Recent data for post-Middle Miocene Siwalik sediments from the eastern and southern Salt Range and the northward adjacent Potwar Basin [14,15] confirm this counterclockwise rotation and show it to be complex in detail. In the Sibi reentrant, a post-Palaeocene clockwise rotation of over 50° relative to the Indian Shield was found in the east-west aligned Loralai Range with absence of rotation in the north-south trending Kirthar and southern Sulaiman Ranges (Fig. 9) [8]. In this paper we extend this rotational analysis to the Trans-Indus and western Salt Range and present quantitative palaeomagnetic evidence for rotational movements of the various range segments of the Mianwali reentrant.

Rotations of individual thrust sheets with respect to the Indian Shield are determined through comparison of observed palaeomagnetic direc-

tions, and notably their declinations, with directions expected in the sampled regions according to the Indian APWP (see [2,4] for discussion).

2. Geological setting

The broad belt of latitudinally aligned subparallel structures that connects the north-south trending Sulaiman Range in the west to the northwest-southeast trending northwestern Himalaya in the east (Figs. 1, 9) was named Hazara Arc by Seeber and Jacob [20]. This typical thrust and fold belt overrides Middle and Upper Tertiary molasse sediments (Murree and Siwalik Formations, Fig. 2), outcropping at the foot of the (Trans-Indus) Salt Range scarp. They are also exposed in the central depression of the Potwar Plateau on the thrust sheet Older sediments, i e. of Early Tertiary, Mesozoic, Permian and Cambrian age, outcrop in the southern scarp and farther northwards within the Hazara Arc. Described as an arc, its structure is more complex in detail, particularly at the Mianwali reentrant which divides the southern scarp into two southwards convex structures (Fig. 1) A widely sinuous eastern segment, spanning the eastern, southern and western Salt Range, extends from the Western Himalayan Syntaxis to the Mianwali reentrant A much smaller western segment extends from the Mianwali reentrant to the Bannu reentrant and comprises the Surghar and the Khisor Range.

The Phanerozoic sequence of the Hazara Arc is detached from basement along the evaporitic Infra-Cambrian Salt Range Formation (Fig 2). The detachment style of tectonics (Fig 1) is clear from lack of relation between mapped surface structures [10,11,21] and abrupt changes in river gradients [22], with seismic zones based on micro-earthquake network data [12,20,23–25] and with gravity data [26]. The prominent scarp of the Salt Range is not visible at all in the Bouguer gravity data [26]. Micro-earthquake data have shown that the MCT-MBT complex continues westwards beyond the Western Himalayan Syntaxis as the IKSZ in the basement. This structure has not been identified in the superstructure other than as an abrupt change in river gradients [22] The surface outcrop of the Murree Thrust (MBT) beyond the northwestern Himalaya follows the western limb

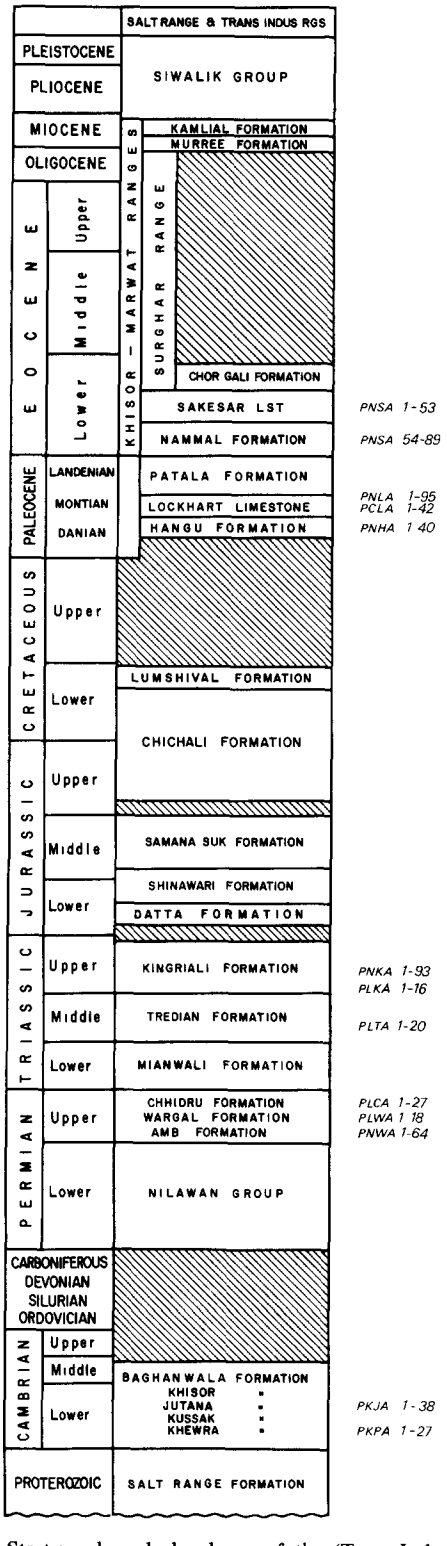


Fig 2 Stratigraphic ideal column of the (Trans-Indus) Salt Range after Ibrahim Shah [28]

of the Western Himalayan Syntaxis into the Hazara Fault (Fig. 1). It cannot be related with the IKSZ nor with another important seismically defined basement fault, the Hazara Lower Seismic Zone (HLSZ) Presence of Infra-Cambrian evaporites accounts for the aseismicity and large width of the zone of detachment which reaches several hundreds of kilometers between the IKSZ and the frontal southvergent structures of the (Trans-Indus) Salt Range scarp.

This scarp of Quaternary origin [27] shows a well exposed Phanerozoic sedimentary sequence, which is unique in its structural position Nowhere along the PFB and the Himalayan Range does such a Phanerozoic sequence outcrop in the external ranges Though the Salt Range sequence does not resemble the external Lower Himalayan sequence, it shows some lithological resemblance with the more internal Tethyan Himalayan sequence and the Kashmir Basin sequence. For reason of its easy access and well exposed sections the (Trans-Indus) Salt Range sequence has been studied intensively and its stratigraphy is well known (Fig 2) [28]. Above the evaporitic Infra-Cambrian Salt Range Formation, locally exposed at the base of the scarp, follows a Lower and Middle Cambrian sandstone shale and dolomite succession which in turn is unconformably overlain by a Permian to Eocene sequence. Apart from a basal tillite and sandstone-shale sequence (Fig. 2, Nilawan Group), this Permian to Eocene sequence consists predominantly of carbonate and sandstone-siltstone. An Upper Cretaceous hiatus represents uplift and erosion prior to the Palaeocene, with subsequent deposition of a Palaeocene/Eocene succession of shallow water carbonates The Oligocene was a regressive period. Syn- and post-tectonic molasse sequences of Miocene and Plio-Pleistocene age (Murree and Siwalik Formation, Fig 2) top the Salt Range succession and outcrop also extensively south of the Salt Range scarp

3. Sampling

Sampling focused on carbonate rather than on sandstone successions for reason of their accurate biostratigraphic control [28]. This policy proved somewhat unfortunate during the analysis as the carbonates had a low to very low magnetic intensity with ubiquitous secondary magnetizations.

Five sections were selected as representative for the various range segments of the Mianwali reentrant, whose structural trends follow the topographical outline of the scarp. From west to east these are (Fig. 1, Table 1): (1) the Saidu Valley in the WNW-ESE trending Khisor Range No representative section could be located in the adjacent N-S trending segment of the Surghar Range, as (2) the Lunda Gorge in the somewhat tectonized bend of the range proved to follow the structural trend of the E-W aligned segment of the Surghar Range, like (3) the Chichali Gorge sampled in the east of the Surghar Range (4) The Dhak Pass section and (5) the Nammal Gorge were sampled as representative for the NNW-SSE trending western Salt Range, which shows local tectonic complications north of the Dhak Pass section.

Whenever possible stratigraphic successions of comparable age were sampled throughout the five sections (Table 1, Figs. 1, 2). Samples were taken as continuous stratigraphic profiles with one core sample drilled per bed, and were oriented with both a magnetic and a solar compass.

(1) Saidu Valley, Khisor Range (32.22°N, 71 00°E). At the foot of the scarp a stratigraphic section of about 25 m was sampled in subhorizontal to slightly NNW dipping fine- to medium-grained crossbedded sandstone, which straddle the top of the Lower Cambrian Khewra Formation (PKPA1–27) into greenish sandy dolomite of the Lower to lower Middle Cambrian Jutana Formation (PKJA1–38)

(2) Lunda Gorge, Surghar Range (32.95°N, 71.20°E). This well accessible and well exposed section north of Mukherwal, is located in the N-S trending segment of the Surghar Range but shows a moderately northern dip characteristic for the E-W trending segment of the range. The tectonized basal part was avoided and short stratigraphic sections were sampled through four Upper Permian to Upper Triassic formations i e.

(a) Thick bedded and somewhat tectonized fossiliferous limestone and dolomite at the top of the Wargal Limestone (PLWA1–18, 40 m section) of Late Permian (Guadelupian) age.

(b) Thin bedded sandy limestone in the overlying base of the Upper Permian (Djulfian) Chhidru Formation (PLCA1–27, 17 m section).

(c) Thin bedded, medium grained and reddish

TABLE 1

Sampling details

Formation, identification	Locality (°N, °E) (Fig 1)	Age	Lithology	Section length (m)	Samples (spec)	Initial intensity (10^{-3} m Am^{-1})	Dip direction (°)	Dip (°)
base Sakesar Limestone/ top Nammal Formation, PNSA1-53/PNSA54-89	32 70, 71 82 (5)	Early Eocene	Nodular limestone/ argillacious limestone	55/57	89 (154)	3-146	52-80	36-39 5 ENE
Lockhart Limestone, PNLA1-95	32 70, 71 82 (5)	Palaeocene (Montian)	bedded and flaggy limestone with shale and marl intercalations	74	95 (168)	16-7776	42-65	26-28 NE
basal Lockhart Limestone, PCLA1-42	33 00, 71 42 (3)	Palaeocene (Montian)		17	42 (66)	29-147	345	42 N
top Hangu Formation, PNHA1-40	32 70, 71 82 (5)	Early Palaeocene	argillaceous sandstone sandy limestone	11	40 (60)	73-5237	35-45	24-26 NE
Kingriali Formation, PNKA1-93	32 70, 71 82 (4)	Late Triassic	porous and more massively bedded dolomite and dolomite limestone	48	93 (137)	3-1468	55-60	47-50 ENE
basal Kingriali Formation, PLKA1-16	32 95, 71 20 (2)	Late Triassic	massively bedded dolomite and dolomitic limestone	6	16 (17)	8-286	37	43 NE
basal Tredian Formation, PLTA1-20	32 95, 71 20 (2)	Middle Triassic	thin bedded sandstone with reddish streaks	24	20 (23)	1085-8024	340	37 N
basal Chhidru Formation, PLCA1-27	32 95, 71 20 (2)	Late Permian (Djulfian)	thin bedded sandy limestone	17	27 (32)	77-1543	10-20	34 N
top Wargal Limestone, PLWA1-18	32 95, 71 20 (2)	Late Permian (Guadelupian)	thick bedded fossiliferous limestone and dolomite	40	18 (31)	23-210	0-25	23-31 N
Wargal Limestone, PNWA1-64	32 70, 71 82 (4)	Late Permian (Guadelupian)	thick bedded fossiliferous limestone and dolomite with chert layers	21/7	64 (105)	3-1201	32 5-37 5	22-35 5 NNE
basal Jutana Formation, PKJA1-38	32 22, 71 00 (1)	late Early Cambrian to early Middle Cambrian	sandy dolomite	12	38 (44)	22-1730	340	6 N
top Khewra Sandstone, PKPA1-27	32 22, 71 00 (1)	Early Cambrian	fine-grained purple sandstone	13	28 (28)	901-342178	340	6 N

coloured sandstone in the basal member of the Middle Triassic Tredian Formation (PLTA1–20, 24 m section)

(d) Massive dolomite and dolomitic limestone at the base of the Upper Triassic Kingriali Dolomite (PLKA1–16, 6 m section), directly above the Tredian Formation

(3) Chichali Gorge, Surghar Range (33 00°N, 71.42°E). Southvergent recumbent folding and thrusting is well developed and exposed at the southern entrance of the Gorge, particularly in the Murree Formation. In the more central part of the gorge a section of partly bedded and partly flaggy limestone at the base of the Palaeocene Lockhart Limestone was sampled over 17 m (PCLA1–42)

(4) Nammal Gorge, western Salt Range (32.70°N, 71 82°E). A total of 64 cores was drilled over two short stratigraphic sections in the Upper Permian (Guadelupian) Wargal Limestone, i e in thick bedded limestone and dolomite at the base of the formation (PNWA1–44, 21 m section) and in thin bedded slightly flaggy fossiliferous limestone (PNWA45–64, 7 m section), with a non-sampled interval of about 70 m Upstream in the gorge, dolomites of a generally rather porous appearance were sampled in the massive basal part of the Upper Triassic Kingriali Dolomite (PNKA1–93, 48 m section). All beds in this gorge section have a moderate northeast dip

(5) Dhak Pass, western Salt Range (32.70°N, 71.82°E). This section is located a few kilometers north of the Nammal Gorge, along the Talagang-Mianwali roadcrossing of the western Salt Range, and shows likewise moderately NE to NNE dipping beds A stratigraphic section of 11 m was sampled in argillaceous sandstone and sandy limestone at the top of the principal reference section [28] of the lower Palaeocene Hangu Formation (PNHA1–40) Another 95 cores were drilled in two gullies throughout the entire succession of the overlying Lockhart Limestone (PNLA1–95, 74 m section) of Palaeocene age, made up of alternating shale, marl and flaggy limestone members. Sampling concentrated on the limestone beds. A further 89 cores were drilled throughout thin to medium bedded argillaceous limestone at the top of the Nammal Formation (PNSA54–89, 57 m section) which contains occasional chert lenses, and through nodular limestone at the base of the conformably overlying Sakesar Limestone (PNSA1–53, 55 m section).

4. Analysis and results

All core samples (570) were sliced into specimens (865) of 2 25 cm height and 2 5 cm diameter, and were washed in diluted HCl prior to measurement of initial remanence in order to remove iron stains acquired upon coring. All measurements were made on a two-axis ScT cryogenic magnetometer interfaced with a HP2100 minicomputer for on-line data reduction Selected specimens, generally one specimen every third sample throughout a profile, were progressively thermally demagnetized in 10–18 steps up to 500–600°C The remainder of the specimens were partially progressively demagnetized in 6–13 steps between 130° and 400–600°C All thermal demagnetizations were continued until the remanent magnetization dropped below the resolution limit of the system (about 3×10^{-3} mA m^{-1}), or until acquisition of viscous magnetization made further measurements impossible (Fig. 4D–F). All measurement and analysis procedures followed standard techniques as described in the accompanying paper [2].

A particular problem with analysis of the carbonate specimens was the non-uniqueness of broad blocking temperature spectra for the various components Identification of mean groupings of individual components and calculation of their mean directions had to be based therefore on analysis of maximum groupings in density distribution projections rather than on similarity in blocking temperature spectra.

Thermal demagnetization showed general presence of four magnetic components, whose interpretation will be discussed in the next section.

(1) Recent field components of both normal and reversed polarity. Some of these components are of pre-tectonic origin and illustrate very recent tectonic activity.

(2) Secondary components of predominantly normal polarity, observed in pre-Lower Tertiary rocks only They are interpreted to be induced during the initial India-Asia collision

(3) Primary components of both normal and reversed polarity were identified in the Cambrian successions, but were absent or of doubtful nature in Upper Permian to Upper Triassic successions. A consistent component in the Lower Tertiary formations could not be identified positively as a

primary or a secondary post-tectonic magnetization. A fold test could not be carried out because of monoclinal sequences, but a primary origin is strongly argued for on circumstantial evidence.

(4) An eastwards and equatorially oriented secondary component of unclear origin was consistently and exclusively observed in the western Salt Range sections

4.1 Saidu Valley

Khewra Formation (PKPA) The samples showed a high intensity of initial remanence (NRM) ranging from $9–10^{-1}$ to 3×10^{2} mA m^{-1} A soft component of minor intensity (about 30% of NRM) and of both normal and reversed polarity was eliminated at 300°C (Figs. 3C, 4A, 5; Table 2 3) The direction of this component is comparable to the present field direction at the sampled locality and is obviously of recent origin Occasionally another hard component of recent origin was eliminated between 500° and 665°C (Fig. 3C). At temperatures above 300°C either a southeastwards and moderately upwards directed component of both normal and reversed polarity (Figs. 3B,C, 5, Table 2· 4,5), or a southwestwards and moderately downwards directed component (Figs 3A,C, 5; Table 2. 6) was eliminated.

Jutana Formation (PKJA) Samples of this formation showed very consistent response to thermal treatment. Below 250–300°C a soft component predominantly of normal polarity and representing less than 60% of initial NRM, was broken down (Figs. 3D, 4A, 4; Table 2· 1) This component is obviously of recent origin At higher temperatures up to 500°C a southwestwards and moderately steep downwards directed component with a predominant reversed polarity became broken down (Figs. 3D, 4A, 5; Table 2: 2). This component is comparable in direction with one of the two harder components observed in the underlying Krewra Formation.

4.2 Lunda Gorge

Wargal Limestone (PLWA) A minority of the specimens showed a soft component of recent origin, which was broken down below 350°C (Figs 4B, 5, Table 2: 11). Most of the samples, however, showed a northward directed component of very shallow inclination, which became broken down below 350°C (Figs 3E, 5; Table 2 12). Some of the specimens showed in addition another northwards but slightly steeper upwards directed component, broken down between 250° and 550°C (Figs. 3E, 5, Table 2: 13).

Chhidru Formation (PLCA) Specimens showed predominant presence of a soft normal polarity component whose direction before bedding correction closely resembles the present field direction (Figs. 3G, 5; Table 2: 9). A recent origin can be argued for, though not convincingly, on basis of the closeness of its non-bedding corrected direction with the present field direction and the consistent presence of this component direction throughout the sampled part of the Lunda Gorge section A secondary origin, (Table 2: 7–9,12,13), however, will be argued for in the next section A minority of specimens showed in addition a harder south-southwestwards and moderately downwards directed component of both normal and reversed polarity (Figs. 3F,G, 5, Table 2: 10)

Tredian Formation (PLTA) and Kingriali Dolomite (PLKA). Both formations showed exclusive presence of a component whose direction (Fig 5; Table 2. 8,9) closely resembles that of the soft component in both the Chhidru Formation and the Wargal Limestone Likewise, equivocal arguments for a post-tectonic recent field origin or a pre-tectonic Early Tertiary origin can be advanced as will be discussed. Clearly, absence of a foldtest in these monoclinal successions hindered conclusive interpretation

4.3 Chichali Gorge

Lockhart Limestone (PCLA) This formation showed predominant presence of a soft component of normal polarity and of recent origin (Figs. 3J–L, 4C, 5; Table 2. 14), which was broken down below 240°C or occasionally above 480°C (Fig 3L) A conjugate recent field component of reversed polarity became broken down between 300° and 400°C (Fig. 3K,L). Another consistently present north-northwestwards and slightly upwards directed component of normal polarity (Fig 3J–L, Table 2: 15) was broken down between 200° and 450°C

TABLE 2

Summary of palaeomagnetic directions and pole positions

No	Component [a]		Mean specimen direction						South pole position				
			D (°)	*I* (°)	*k*	α_{95} (°)	*N*	polarity (N/R)	lat (°S)	long (°E)	*dp* (°)	*dm* (°)	palaeolat (°N)
A Saidu Valley													
1	Rnr	PKJA	335	51	16	6 5	33	(31/2)					
2	Pnr	PKJA	202 5	53 5	9	11	21	(5/16)	20 5	51	11	15 5	34 5
3	Rnr	PKPA	0 5	39 5	22	8 5	14	(9/5)					
4	Sn	PKPA	328 5	22	17	14	8	N	54 5	132 5	7 5	14 5	11 5
5	Sr	PKPA	135	−28 5	31	14	5	R	45 5	149	8 5	15	15
6	Pr	PKPA	203	31 5	40 5	8	9	R	36	43 5	5	9	17 S
B Lunda Gorge													
7	Sn [b]	PLKA	346 5	2	14	18 5	6	N	55 5	95 5	3	6	1
			(354 5	41 5)									
8	Sn [b]	PLTA	354	5	29	6 5	19	N	59	82	3 5	6 5	2 5
			(357	41 5)									
9	Sn [b]	PLCA	3	8 5	29	5 5	25	N	61	65	3	5 5	4 5
			(357	41	31	4 5	32)						
10	Prn?	PLCA	150	45	10	20	7	(4/3)	24	100 5	15 8	25	26 5
11	Rn	PLWA	345 4	48 5	39 5	20	3	N					
12	Sn1	PLWA	353 5	−2 5	56 5	4	26	N	55	82 5	2	4	1 5 S
13	Sn2	PLWA	5 5	−12	23	10 5	10	N	50 5	62	5 5	10 5	6 S
C Chichali Gorge													
14	Rn?	PCLA	354 5	40	94	2	51	(50/1)	55	84 5	1	2	1 5 S
	(Sn?		352 5	−2 5	97	2	45	N)					
15	Pn	PCLA	335	−13	11	8 5	28	N	44	107	45	9	6 5 S
D Nammal Gorge													
16	Rnrl	PNKA	0	44	24 5	4	48	(46/2)					
17	Rnr2 [c]	PNKA	2	47 5	19	14	7	(5/2)					
18	Sn	PNKA	0 5	−9	11 5	14	11	N	53	72	7	14	4 5 S
19	Qun	PNKA	71	5 5	9	19	8	N					
20	Rn1	PNWA	346 5	50	42	10 5	6	N					
21	Rnr2 [c]	PNWA	351 5	53 5	16	14	8	6/2					
22	Sn	PNWA	357	9 5	15	5 5	52	N	62	78 5	2 5	5 5	5
23	(Pn?	PNWA	347 5	−28	10 5	6	57	N	41	88	3 5	6 5	15 S)
24	Qur	PNWA	262	19	16	12 5	10	R					
E Dhak Pass													
25	Rn1	PNSA	4 5	51 5	30	5	27	N					
26	Rn2 [c]	PNSA	5	46 5	28	4 5	40	N					
27	Rr2 [c]	PNSA	168 5	−50 5	12 5	10	19	R					
28	Pn	PNSA	9	−3	13 5	9 5	20	N	55	55	4 5	9 5	1 5
29	Qun	PNSA	89 5	−7	10	14 5	12	N					
30	Rn1	PNLA	5 5	49	21	3 5	78	N					
31	Pnr	PNLA	18	−10 5	12	8	30	24/6	48 5	44	4	8	5 S
32	Qun	PNLA	84	−1	9	11	22	N					
33	Rn1	PNHA	352	49	40	9	8	N					
34	Rn2 [c]	PNHA	9 5	47	22 5	10	11	N					
35	Rr2 [c]	PNHA	193	−37 5	18	10	13	R					
36	Prn	PNHA	173	−2 5	5	14	26	(7/19)	58	85	7	13 5	1
37	Qurn	PNHA	86 5	0	8 5	16 5	11	(7/4)					

D = declination, I = inclination, k = precision parameter [55], α_{95} = half-angle cone of 95% confidence, N = number of specimens, dp, dm = half-angles of the oval of 95% confidence about the pole position

[a] See caption to Fig 3 R = recent field component, S = secondary component, P = primary component, n = normal polarity, r = reserved polarity, qu = component of uncertain origin

[b] Directions between brackets are before bedding correction, the possibility of a recent field origin cannot be excluded

[c] Recent field component corrected for bedding Present field direction $D = 1°$, $I = -49°$

4 4 *Nammal Gorge*

Wargal Limestone (PNWA) Specimens from this formation showed a conspicuous combination of a soft component eliminated below 300°C to occasionally 400°C (Fig. 5, Table 2: 22) and a harder component eliminated between 250° and 450°C (Figs 3M,O, 5, Table 2· 23) Both components are of normal polarity and show good grouping directions. In a minority of specimens soft components of a predominant normal polarity were found which were acquired prior to (Fig 5; Table 2 21) or after recent tilting of the beds (Figs 3M–O, 5, Table 2: 20) In addition an

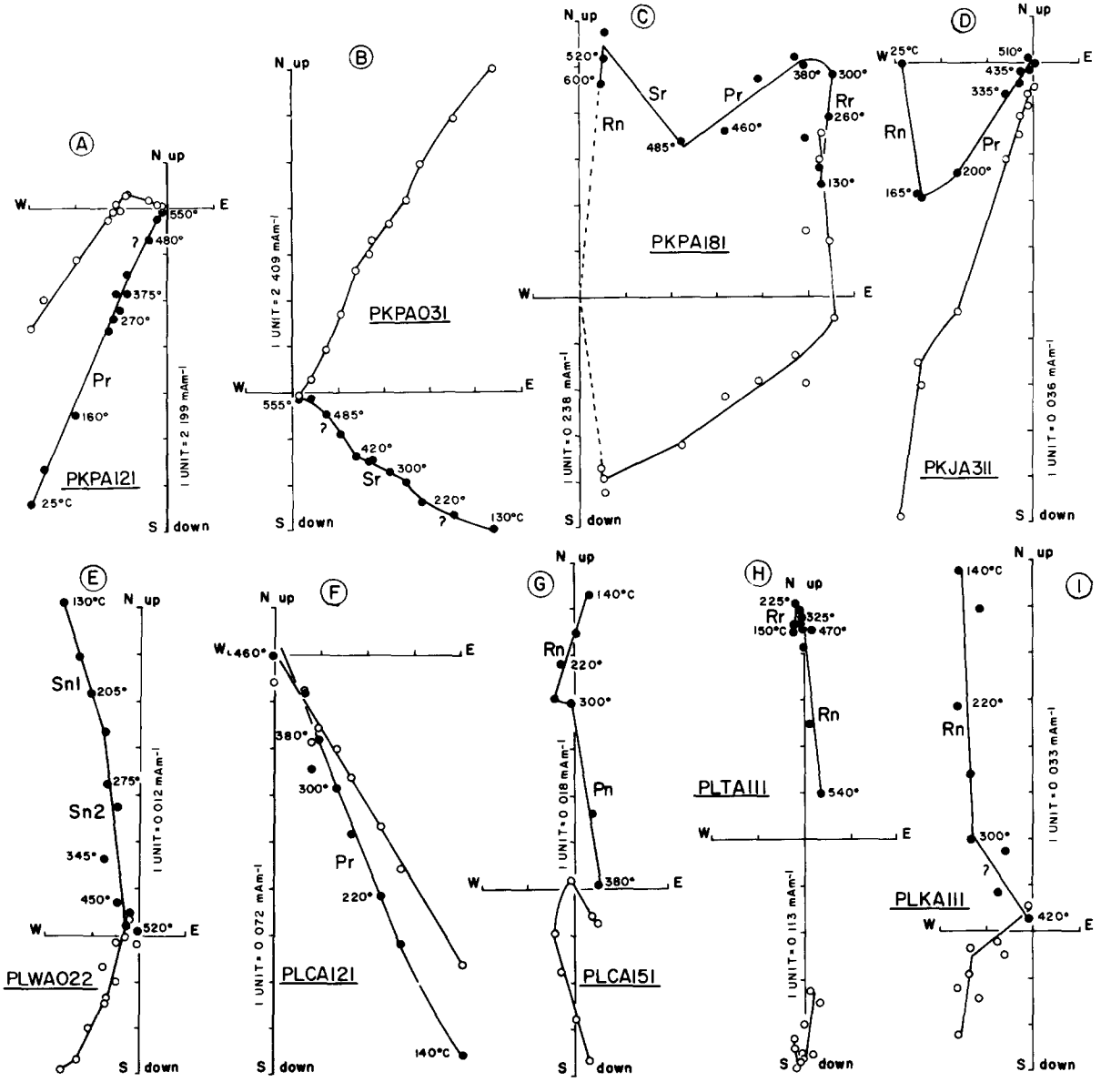
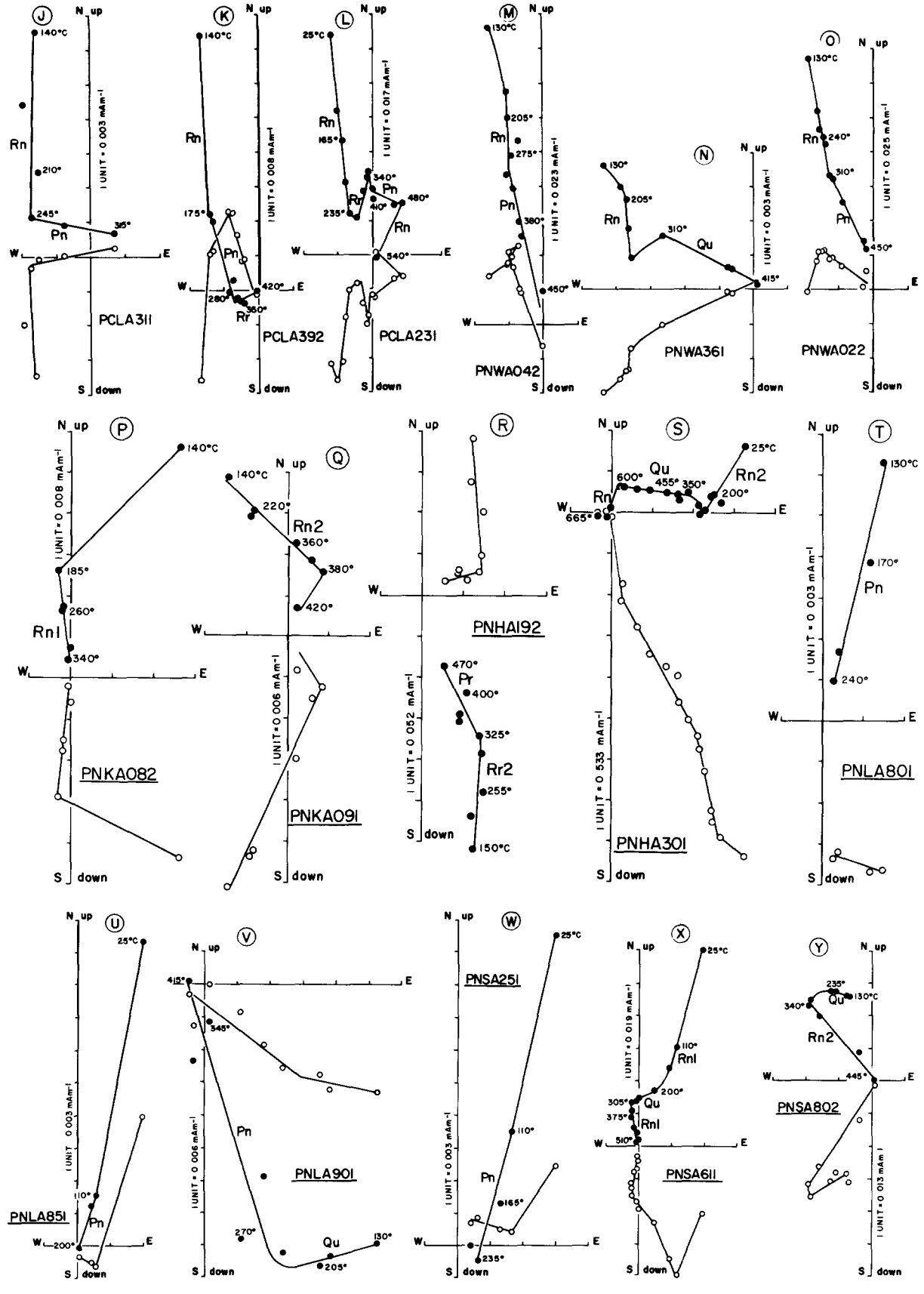


Fig 3 Zijderveld [45] diagrams of some representative specimens during thermal demagnetization The circles indicate successive positions—in orthogonal projection—of the end points of the resultant magnetization vector during progressive thermal demagnetization Open circles indicate projections on the vertical east-west plane, and dots indicate projections on the horizontal plane Numbers denote successive peak temperature values R = recent field component, S = secondary component, P = primary component, n = normal polarity, r = reversed polarity, Qu = component of uncertain interpretation


J
N up
140°C
Rn
210°
245°
Pn
365°
W
E
S down
PCLA311
1 UNIT = 0 003 mAm⁻¹
K
140°C
Rn
175°
Pn
280°
Rr
360°
420°
PCLA392
1 UNIT = 0 008 mAm⁻¹
L
25°C
Rn
165°
Pr
340°
Pn
480°
235°
540°
PCLA231
1 UNIT = 0 017 mAm⁻¹
M
130°C
205°
Rn
275°
Pn
380°
450°
PNWA042
1 UNIT = 0 023 mAm⁻¹
N
130°
205°
Rn
310°
Qu
415°
PNWA361
1 UNIT = 0 003 mAm⁻¹
O
130°C
Rn
240°
310°
Pn
450°
PNWA022
1 UNIT = 0 025 mAm⁻¹
P
140°C
185°
260°
Rn1
340°
PNKA082
1 UNIT = 0 008 mAm⁻¹
Q
140°C
220°
Rn2
360°
380°
420°
PNKA091
1 UNIT = 0 006 mAm⁻¹
R
PNHA192
470°
Pr
400°
325°
Rr2
255°
150°C
1 UNIT = 0 052 mAm⁻¹
S
25°C
600°
Qu
455°
350°
Rn2
200°
Rn
665°
PNHA301
1 UNIT = 0 533 mAm⁻¹
T
130°C
170°
Pn
240°
PNLA801
1 UNIT = 0 003 mAm⁻¹
U
25°C
110°
Pn
200°
PNLA851
1 UNIT 0 003 mAm⁻¹
V
415°
345°
Pn
PNLA901
270°
Qu
130°
205°
1 UNIT = 0 006 mAm⁻¹
W
25°C
PNSA251
110°
Pn
165°
235°
1 UNIT = 0 003 mAm⁻¹
X
25°C
1 UNIT = 0 019 mAm⁻¹
110°
Rn1
200°
305°
Qu
375°
Rn1
510°
PNSA611
Y
235°
340°
Qu
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445°
PNSA802
1 UNIT = 0 013 mAm⁻¹

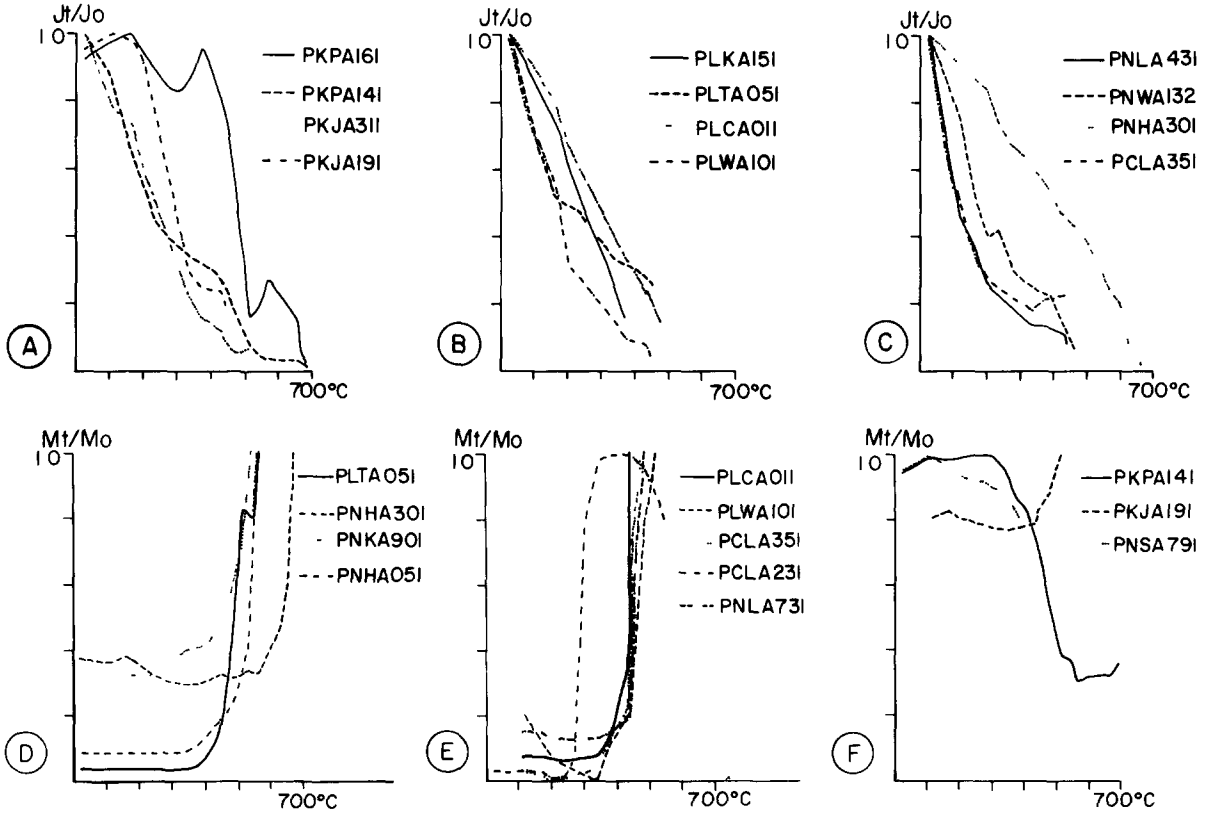


Fig 4 Representative normalized curves showing the decay of remanent magnetization (A–C) and changes in susceptibility (D–F) during thermal treatment

eastwards directed component of low inclination was observed in a minority of the specimens (Figs. 3N, 5; Table 2 24). This component was observed throughout all formations sampled in the western Salt Range sections (Dhak Pass, Nammal Gorge), but was not identified in the three sections sampled in the Khisor and Surghar Ranges.

Kingriali Dolomite (PNKA) Specimens from this formation showed poor magnetic stability and the intensity of magnetic remanence was close to the measurement limit of the magnetometer The identifiable magnetic content of the specimens was dominated by a component of recent (Figs. 3P,Q, 5, Table 2: 16,17) and mainly post-tectonic origin (Figs. 3P, 5, Table 2: 16). A minority of specimens showed a soft component of shallow northern dip (Fig. 5; Table 2. 18), whose direction is comparable to the predominant soft component observed in the Wargal Limestone (Fig. 5, Table 2 22) The peculiar shallow eastwards directed component was likewise observed in a limited number of specimens (Fig 5, Table 2 19).

4 5 *Dhak Pass*

Hangu Formation (PNHA) Specimens from this formation showed a characteristic southwards and equatorially directed component of predominantly reversed polarity (Figs. 3R, 5, Table 2 36) and hard magnetic properties, broken down between 200° and 500°C In addition soft recent field components of both pre-tectonic (Figs. 3R,S, 5; Table 2. 34,35, N and R polarities) and post-tectonic origin (Figs 3S, 5, Table 2 33) were present, and also a rather dispersed group of directions (Figs. 3S, 5; Table 2: 37) comparable to the eastwards and equatorially directed component found throughout the studied sections of the western Salt Range.

Lockhart Limestone (PNLA). The upper part of

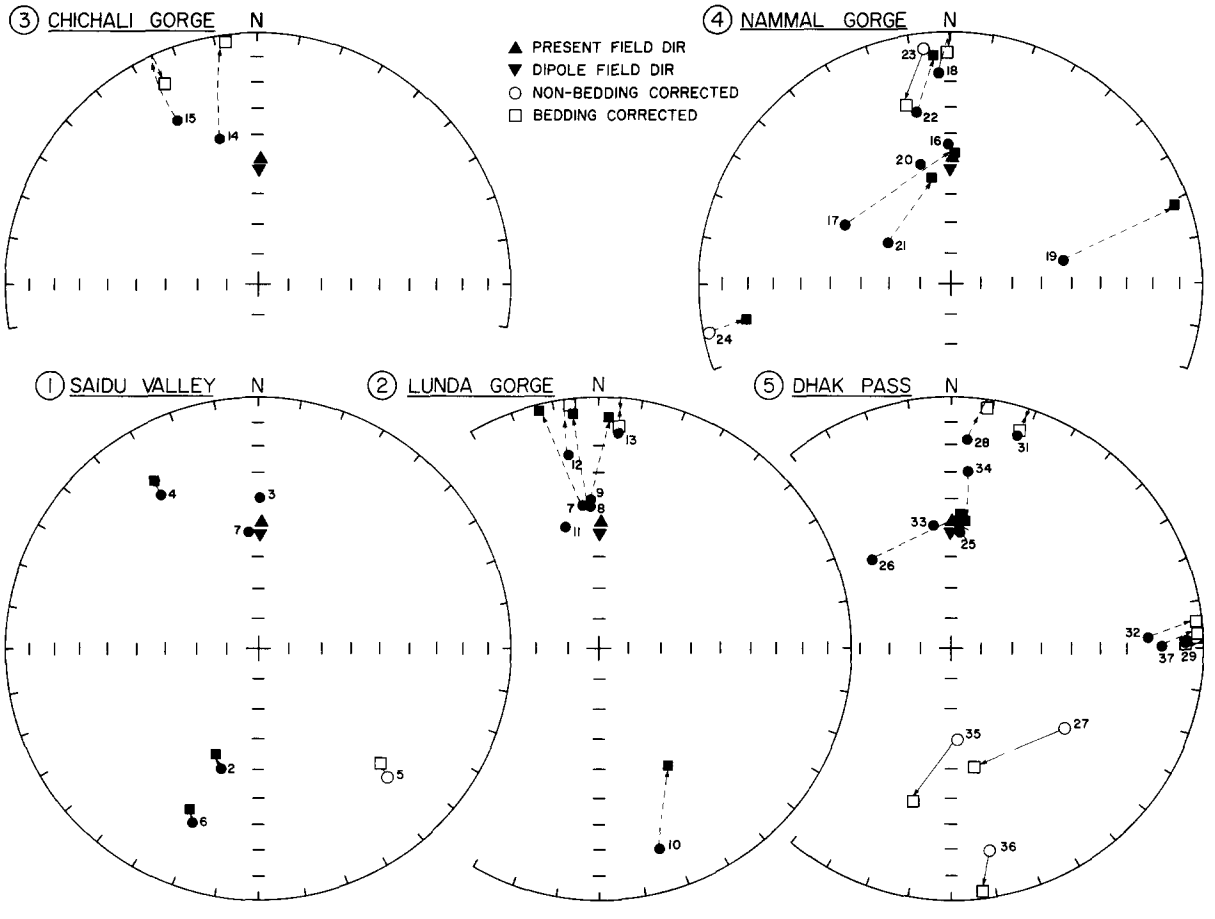


Fig 5 Equal area projection of mean results (Table 2), before (circles) and after (squares) correction for bedding Open/closed symbols indicate upper/lower hemisphere projection

the sampled succession showed exclusive presence of a recent field component of normal polarity (Fig. 5; Table 2: 30). Specimens from the basal part, in contrast, showed a northwards and slightly upwards directed component of normal polarity, eliminated below 350°C to occasionnally 450°C (Figs 3T–V, 5; Table 2. 31). The shallow eastwards directed component (Figs 3V, 5, Table 2· 32) was also present

Sakesar Limestone and Nammal Limestone (PNSA) Like the underlying Lockhart Limestone, three components were observed in this succession, i.e.: a predominant recent field component of both pre-tectonic (Figs. 3Y, 5, Table 2. 26,27, N and R polarities) and post-tectonic origin (Figs. 3X, 5, Table 2: 33), a northwards and equatorially directed component (Fig. 5, Table 2 28), and the eastward directed component of shallow inclination characteristic for the western Salt Range (Figs. 3X,Y, 5, Table 2 29)

5. Interpretation

Recent field components were identified from comparison of their directions, on occasion after correction for bedding, with the local field direction. Other components were interpreted from comparison of bedding corrected directions and corresponding pole positions with directions expected in the sampled region according to the Late Cretaceous and Tertiary segment of the Indian APWP, according to newly obtained Permian and Triassic data from Gondwana formations of the Damodar Valley in northern India [29,30], and

according to Cambrian data from the eastern Salt Range [16,17]. Rotational movements relative to the Indian Shield as expected from moulding of structural trends along the reentrant were accounted for in this comparison (Fig. 6).

Several well identifiable components could not be interpreted unambiguously in absence of a fold test in monoclinal sections. It was not possible to decide conclusively between a non-bedding corrected recent field origin (Fig. 5, Table 2 7–9,22,28,31), and either a primary origin (Fig. 5, Table 2· 28,31) or a Late Cretaceous/Early Tertiary secondary origin (Fig 5, Table 2· 7–9,22) The latter may be attributed either to a Late Cretaceous/Early Tertiary period of uplift and erosion as observed in a palaeomagnetic study of northern Baluchistan [8], or to thermochemical effects associated with the initial India-Asia collision [5]. A recent field origin proved to be less likely, as bedding corrected directions for such components fitted well within the framework of unambiguously interpreted palaeomagnetic data and supported the pattern of rotations relative to the Indian Shield as expected on structural grounds along the reentrant

5 1. Saidu Valley

The harder southwestwards and moderately downwards directed components in the Early Cambrian Khewra Sandstone and the late Early to early Middle Cambrian Jutana Formation (Fig. 5, Table 2 2,6) are interpreted as primary magnetizations. Their directions are close to those expected in the southern Khisor Range on the basis of primary components of similar age observed in the Khewra Sandstone and Baghanwala Formation of the southern Salt Range [16,17]. The southeastward directed component in the Khewra Sandstone closely resembles Early Tertiary Indian directions (Fig 6· 1) and is interpreted as a secondary magnetization. Both the primary Cambrian and the Early Tertiary secondary direction show a counterclockwise offset over about 15° with respect to the southern Salt Range data and the Indian APWP respectively (Fig. 6 1). The southern Salt Range in itself has undergone a counter-

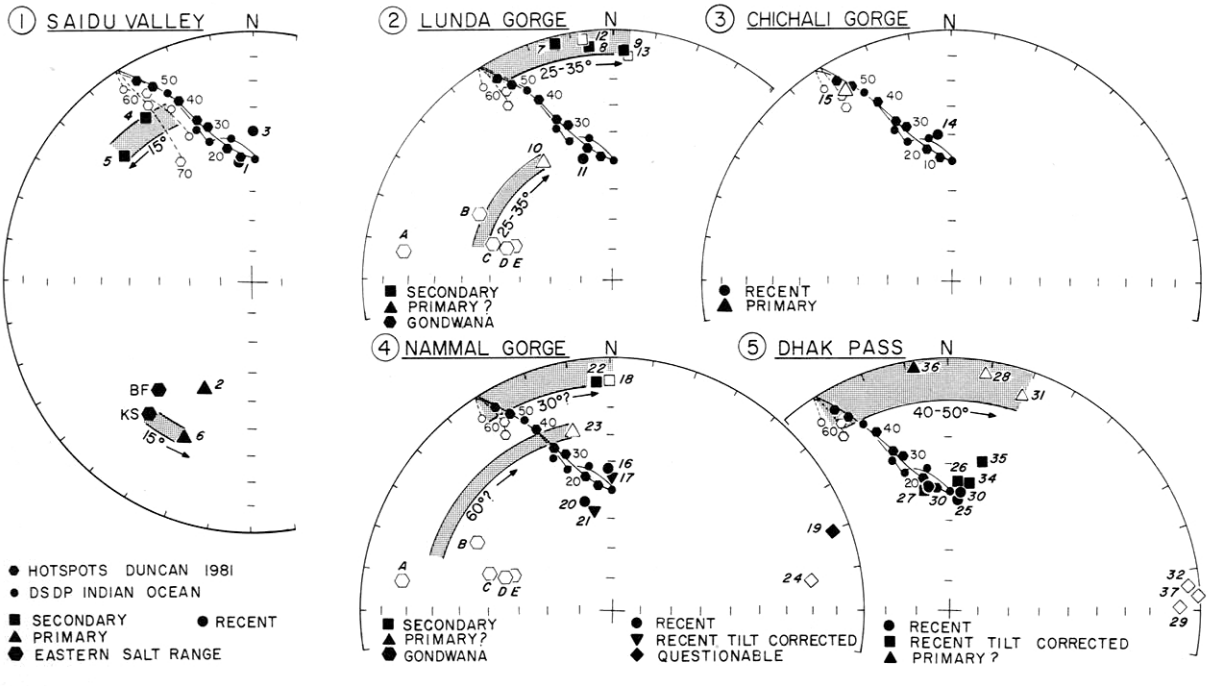


Fig 6 Comparison of mean directions observed for the primary, secondary, recent field and questionable components with directions expected at the five localities according to simulated Tertiary APWP's for the Indian plate (a and b, see text) Italic numbers correspond with results listed in Table 2 Other numbers indicate ages in Ma or magnitudes of rotation relative to the Indian plate *A–E* indicate newly available Permo-Triassic palaeomagnetic data from Indian Gondwana formations [29,30]

clockwise rotation over 20–30° relative to the Indian Shield [14,19]. The above observation indicates thus a total counterclockwise rotation of the sampled part of the Khisor Range over 35–45° relative to the Indian Shield, which may have been initiated before about 40 Ma ago as suggested by the about 15° discrepancy between the Early Tertiary secondary components and the Indian APWP

5 2 Lunda Gorge

Absence of a fold test in this moderately northwards dipping succession prevented unambiguous interpretation in terms of either a non-bedding corrected recent field component or a bedding corrected Early Tertiary secondary component for soft and directionally similar components in the Upper Permian Chhidru Formation, the Middle Triassic Tredian Formation and the Upper Triassic Kingriali Formation (Table 2: 7–9; Figs. 5,6: 2). A recent field origin can be excluded, in contrast, for the directionally close softer component in the underlying Upper Permian Wargal Limestone (Table 2 12, Figs. 3E, 5). This makes it most likely that the above components are likewise of an Early Tertiary secondary origin. These secondary components indicate about 25–35° of clockwise rotation of the sampled part of the Surghar Range with respect to the Indian Shield (Fig. 6 2). Comparison of the harder southwards and downwards directed component in the Chhidru Formation (Fig 5; Table 2: 10), whose upwards inclination definitely indicates a primary origin, with Permo-Triassic directions from Indian Gondwana formations (Fig. 6: 2, AGE) indicates a clockwise rotation over about 50° This 15–25° difference in rotation between the Early Tertiary and Late Permian data supports the conclusion reached from the Saidu Valley data that movements relative to the Indian Shield have occurred before the Early Tertiary, probably during the initial stage of the India-Asia collision

5 3 Chichali Gorge

The well determined north-northwestward and slightly upward directed component in the Palaeocene Lockhart Limestone (Table 2: 15; Fig 6: 3), is in good agreement with directions expected from the Early Tertiary Indian APWP (Fig 6. 3). A primary origin and absence of rotation of the sampled part of the Surghar Range relative to the Indian Shield is concluded.

5 4 Nammal Gorge

The soft northward and equatorially directed components in the Wargal Limestone and Kingriali Dolomite (Table 2 22,18; Figs 5,6. 4) are comparable to secondary components observed in the Upper Permian to Upper Triassic sediments from the Lunda Gorge (Table 2· 7–9; Figs. 5,6 2). An Early Tertiary origin is likewise concluded. Comparison with the Early Tertiary segment of the Indian APWP indicates a clockwise rotation over about 30° relative to the Indian Shield. The harder component in the Wargal Limestone, whose upwards pointing direction is of an obvious pre-Tertiary and most probably a primary origin (Figs. 5,6 4, Table 2 23), also indicates a clockwise rotation when compared with newly available Permo-Triassic data from Indian Gondwana formations (Fig. 6 4, A–E). Its magnitude of about 60° exceeds that concluded from the Early Tertiary data and further supports the conclusion reached above from analysis of the Saidu Valley and Lunda Gorge section for pre-Early Tertiary movement relative to the Indian Shield. No sensible explanation can be offered for the eastward and equatorially directed components found in the Nammal and the Chhidru Formation (Figs 5,6: 4, Table 2: 19,24), neither for similar directions in the Tertiary formations from the northward adjacent Dhak Pass section (Figs 5,6. 5, Table 2: 29,32,37).

5 5 Dhak Pass

The equatorial inclination of the northward directed components in the three Lower Tertiary formations (Fig 5,6· 5) agrees with the expected inclinations for that period (see also [2,31]) These components are interpreted to be of primary origin. Comparison of their declinations with the Indian APWP shows a clockwise rotation over 40–50° and supports the conclusion reached from the southward adjacent Nammal Gorge data

6. Discussion

Two general observations can be drawn from the pattern of rotational deformations in the Hazara Arc detachment.

(1) Palaeozoic and Mesozoic rocks indicate

larger rotations than Lower Tertiary rocks.

(2) Magnitude and sense of rotations follow the changing structural trend along the reentrant, the Lunda Gorge data excepted

The first observation implies a pre-Early Tertiary phase of rotational movement relative to the Indian Shield. A similar conclusion was reached in a palaeomagnetic study of the Kashmir Basin [1]. The Kashmir data showed a disagreement in rotation relative to the Indian Shield between on the one hand secondary components of Late Tertiary age and on the other hand primary components of Early Palaeozoic and Late Palaeozoic/Early Mesozoic age. This points to a pre-Late Tertiary phase of rotational movement. It was not possible, however, to conclude from the Kashmir data whether this movement was confined to the suprastructure of the Panjal Nappe or that basement was involved In the present case of the Mianwali reentrant, the pattern of changes in sense and magnitude of rotations along with the changing structural trend of the reentrant makes it more likely that the pre-Early Tertiary phase was confined to the suprastructure These early rotations vary from 20 to 30° clockwise in the southern Khisor Range (N.B comparison is made in Fig. 6. 1 with data from the southern Salt Range, rotation of the latter with respect to the Indian Shield (Fig 7) is not included) to 20–30° counterclockwise in the western Salt Range [4,19] (Fig. 6) and show thus the same sense of rotation as the post-Early Tertiary rotation phase. They conform to expectations based on changes in structural trend along the reentrant. It seems a reasonable assumption that moulding of structural trends of the detached suprastructure of the Hazara Arc along the Mianwali reentrant was initiated during this pre-Early Tertiary phase and that moulding became more pronounced during the post-Early Tertiary phase. The presence of recent field-like components of both pre- and post-tectonic origin exemplifies the recent nature of the latter tectonic phase, whose Plio-Pleistocene origin [27] is conclusively supported by rotations observed from primary magnetization components in the Upper Siwalik sediments [14,15] This tectonic phase is thus comparable in timing and is probably related to Plio-Pleistocene oroclinal bending of the Himalayan Arc [4] and development of intramontane basins throughout the Western Himalayan Syntaxis and Himalayan Arc [13,32,33] It is not clear what tectonic event actually caused the hypothetical pre-early Tertiary phase of rotational movement. Initial contact at about the Creta-

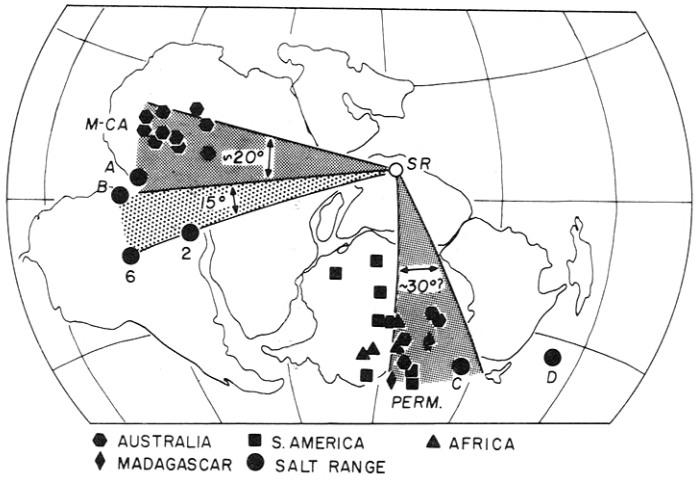


Fig 7 Comparison of Middle Cambrian (M-Ca) and Permian palaeomagnetic poles from Gondwana continents reassembled in the Smith and Hallam [46] reconstruction modified after Griffith's [47] with Early to early Middle Cambrian (*A*,*B* [16,17]) and Early Permian (*C*,*D* [18]) data from the southern Salt Range, and Early to early Middle Cambrian data from the Khisor Range (2,6 see Table 2 2,6) The Salt Range data indicate a 20–30° counterclockwise rotation relative to the Gondwana continent and the Indian Shield as noted before [19] The Khisor Range data show a further counterclockwise rotation over about 15°

ceous/Tertiary time boundary or shortly thereafter between northwestern Greater India and an Andean Arc bordering southern Asia may have induced this rotational movement. Alternatively, it may be related to obduction of the Kohistan Complex along the Main Mantle Thrust (MMT) for which isotopic age determinations vary between 67 and 85 Ma [34–38].

The combined rotations for the pre-Early Tertiary and the Plio-Pleistocene phase (Fig. 8) vary with the structural trends from a 35–45° counterclockwise rotation in the NE-SW trending Khisor Range via absence of notable rotation in the E-W trending segment of the Surghar Range to an about 60° clockwise rotation in the NNW-SSE trending western Salt Range. Data from the southern Salt Range [14–19] indicate an up to 35° counterclockwise rotation in the ENE-WSW trending southern Salt Range, which seems locally complex when more detailed data are available Reconstruction of the individual range segments to their pre-Tertiary position with respect to the Indian Shield (Fig. 8), shows an original latitudinal alignment of the southern part of the Hazara Arc. This arc has arrived at its present position and structure during a probable two-phase movement. Southwards directed counterclockwise rotational movement over 20–45° around an Eulerian pole in the Western Himalayan Syntaxis east of the Salt Range resulted in more complex rotational deformation in the region of Mianwali reentrant. Here movement was obstructed by the NNW trending basement structure of the Sarghoda Ridge (Fig 1), formed of Upper Tertiary molasse [9,39]. Boundary faults of this ridge are still seismically active [12,27] (Fig 1).

Rotation data from the Lunda Gorge section (Figs. 1, 6.2, 8), which is located close to the changeover of the north-south outline of the Surghar Range into an east-west strike, does not follow this regular pattern of moulding of structural trends around a basement obstruction. These data indicate a 30–50° clockwise rotation relative to the Indian Shield (Fig. 6. 2), which is neither expected from its location at the northern end of the north-south trending segment of the Surghar Range, nor from the prevailing east-west strike of the beds. Obviously, the position of the Lunda

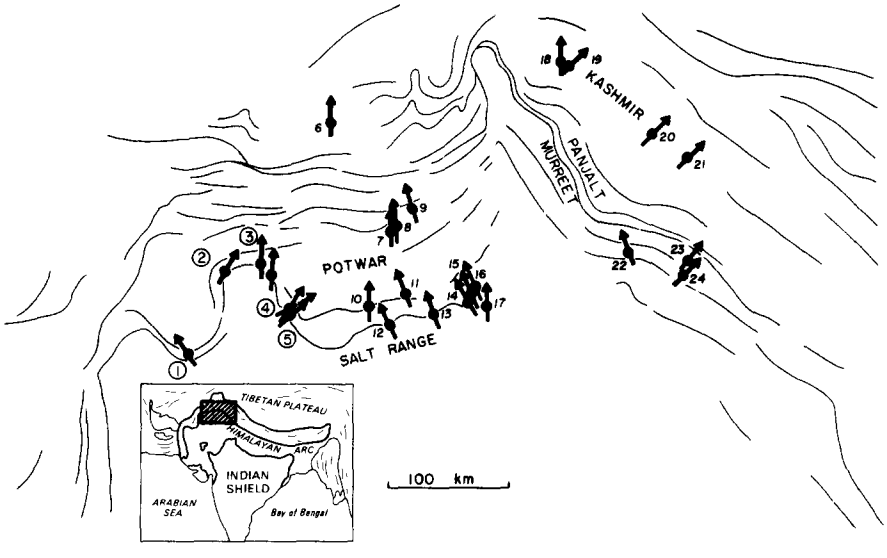


Fig 8 Relation between structural trends in the Hazara Arc and the Western Himalayan Syntaxis after Calkins et al [11] and "palaeomagnetic rotation vectors" whose azimuth indicates the magnitude of rotation of the tectonic units with respect to the Indian Shield Encircled numbers indicate present results, *1* = Saidu Valley, *2* = Lunda Gorge, *3* = Chichali Gorge, *4* = Nammal Gorge, *5* = Dhak Pass, *6* = Dhaud Khel [14], *7* = Hasal Kas [14,15], *8* = Ratha Kas [14,15], *9* = Kaulial Kas [14,15], *10* = Chinji-Nagri [14], *11* = Chakwal Bhaun [14], *12* = Nilawan-Sardahi Gorge and Nurpur-Kushab road [18], *13* = Khewra Gorge and road section [16,17], *14* = Kotal Kund [14], *15* = Dhala Nala [14], *16* = Tatrot [14], *17* = Jalalpur [14], *18*, *19* = northwestern Kashmir [1], *20* = Khrew and Bren Spur [1], *21* = Liddar Valley [1], *22* = Kalakot [2], *23* = "lower" Murree [2], *24* = "lower Upper" Murree [2], *25* = Nowshera reef [48] Compare Fig 1 for location

Gorge section close to the knickpoint of the Surghar Range has rendered the section less suitable for determination of rotational movement of the Surghar Range. The obtained results presumably reflect locally complex movements

The pronounced moulding of structural trends along the Mianwali reentrant has a more extensive but less pronounced counterpart in the NW-SE trending Sibi reentrant of northeastern Baluchistan. This reentrant is situated between the N-S trending Kirthar Range and the E-W trending Loralai Range of the Sulaiman lobe (Fig 9) Palaeomagnetic data from both regions [8] strongly support structural arguments for obstruction by basement topography of thrust sheets moving towards the foreland. Both reentrants are located at the northwestern end of NW-SE trending basement ridges, i e. the Sarghoda Ridge and the Jacobabad-Khairpur Ridge (or Sukkur High), which formed positive structures during the Late Mesozoic and the Early Tertiary. During the Late Tertiary, inversion of relief occurred at the northwestern end of the highs in the present reentrant regions, where massive obstructional molasse wedges of post-orogenic Murree and Siwalik sediments [6,9] were formed.

Whereas the Mianwali reentrant represents a pronounced, but relatively minor, structure formed internally within a large thrust sheet detached along the evaporitic Salt Range Formation, it has been suggested that the larger Sibi reentrant resulted from its position at the very fringe of such an evaporitic detachment [6] The suggested presence of an Infra-Cambrian evaporitic horizon beneath the Sulaiman lobe, petering out in the reentrant region, is still to be confirmed by drilling. A similar mechanism proposed by Seeber et al [12] for the extremely tight Western Himalayan Syntaxis is supported by documented presence of the evaporitic Infra-Cambrian Salt Range Formation beneath the Hazara Arc and absence or sporadic occurrence of evaporites beneath the thrust sheets of the northwestern Himalaya [40,41] Newly available palaeomagnetic data from thrust sheets at both limbs of the Syntaxis, i.e the Panjal Nappe [1] and the Riasi thrust [2] in the east and the Salt Range–Potwar Plateau [14,15] in the west confirm this tweezer-like movement concept (Figs 8, 9).

Reconstruction of the original position of the allochthonous suprastructure of the Western Himalaya Syntaxis, for a 20–45° counterclockwise rotation of the Salt Range–Potwar Plateau structural unit and a 45° clockwise rotation of the northwestern Himalayan region, aligns the fronts of the thrust sheets along east-west axes The above palaeomagnetic data thus support Crawford's [3] hypothesis that the Salt Range had been rotated over about 75° counterclockwise from an original position in line with the main Himalayan front around a pole near the eastern end of the Range, in the sense that one has to read this 75° rotation as relative to the allochthonous northwestern Himalaya and not relative to the Indian Shield. Crawford's suggestion implies that the southwards thrust movement of the Hazara Arc exceeded that of the northwestern Himalayan thrust sheets by an amount comparable to the present distance between the Salt Range front and the KLSZ. This distance of about 200 km falls well below the resolution of the palaeomagnetic data. It will have to be established by other means (i.e deep seismic sounding and balanced cross sections [42]), whether the present distance between the thrust fronts at the eastern and western limb of the Syntaxis results from a much larger evaporite facilitated thrust movement in the west or merely represents southwards progradation of thrust fronts with smaller thrust movements

The secondary components observed in the pre-Lower Tertiary formations can be dated around 55 Ma from intersection of the loci of their poles with the Indian APWP (Fig 10A. 7–9,12,13,18,22) Comparable ages have been concluded for secondary components widely observed in the internal Himalayan region, i e in the eastern Hindu Kush, Ladakh and in central Nepal (Fig. 10B), which are attributed to thermo-chemical effects associated with the initial India-Asia collision [5]. Such a process also may have induced the here observed secondary components. Alternatively, these components could have been induced during uplift and erosion on an outer rise [43,44] on the subducting continental crust This axial bulge prograded southwards leading to Late Palaeocene/ Early Eocene acquisition of secondary components in the further southwards located southern Sulaiman and northern Kirthar Ranges (Fig 10A: A–F)

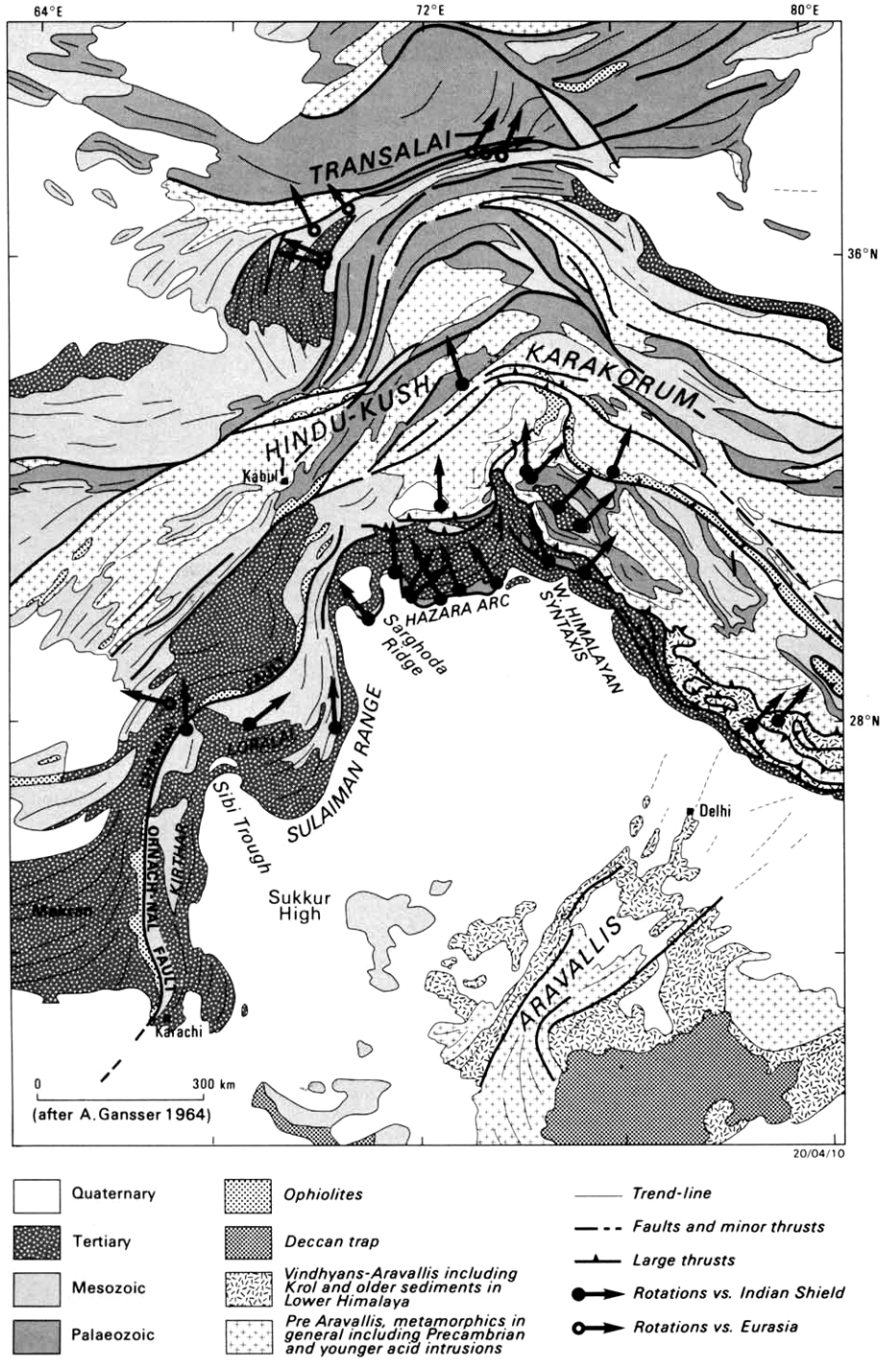


Fig 9. Overview of magnitude and sense of rotation of thrust sheets versus the Indian Shield (full dots) or Eurasia (open dots) Data according to [49] (Outer Pamir, Transalai), [19] (Outer Pamir, Transalai, Salt Range), [50,51] (Ladakh), [52] (Krol Belt), [1] (Kashmir), [2] (Jammu), present paper (Trans-Indus Salt Range), [14] (Salt Range–Potwar Plateau), [8] (northeastern Baluchistan), [47,53,54] (Nowshera reef, Chitral, Spinatizha Complex)

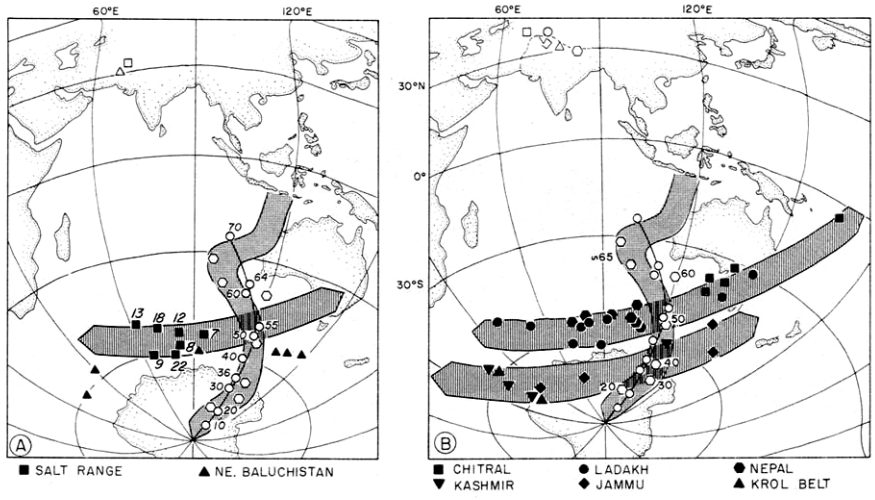


Fig 10 A Comparison of Indian APWP's compiled from DSDP data and Deccan Trap data (hexagons) and poles simulated from India-Asia relative movement data fixed to a hotspot frame in the Atlantic Ocean ([2,4], open circles, italics indicate ages in Ma) with secondary pole positions from the Mianwali reentrant (Table 2) and with secondary components obtained from a study in northeastern Baluchistan [8] The loci of the secondary pole positions from the Mianwali reentrant intersect the Indian APWP's between about 50 Ma and less than 60 Ma The Baluchistan secondary poles appear to be younger B Secondary pole positions obtained from the Himalayan region and the Eastern Hindu Kush group into two swaths An older swath, comparable to that for the Mianwali reentrant (A), is made up by data from the more internal zones These result probably from thermo-chemical effects associated with the initial India-Asia collision The younger swath is made up of data from the more external zones which probably reflect thrust movements towards the Indian Shield

7. Conclusions

Comparison of palaeomagnetic results from five sections along the Mianwali reentrant with the Indian APWP prove the secondary origin of the reentrant Magnitude and sense of rotations closely follow changes in structural trends Rotations indicated by Palaeozoic and Mesozoic formations exceed by about 20° rotations shown by Lower Tertiary formations. This indicates a two-stage deformation process with an initial pre-Early Tertiary moulding phase and a Plio-Pleistocene main tectonic phase The initial phase may be related either to Late Cretaceous obduction of the Kohistan Complex or to initial contact between Greater India and Southern Asia at about the Cretaceous/Tertiary time boundary.

The particular U-shape of the Mianwali reentrant has resulted most probably from moulding of the southwards moving and counterclockwise rotating Hazara Arc thrust sheet around the basement topography of the Sarghoda Ridge

Contrasting rotations of thrust sheets at opposite limbs of the Western Himalayan Syntaxis, i e a general 20–45° counterclockwise rotation of the Hazara Arc thrust sheet in the west and an about 45° clockwise rotation of the Panjal Nappe, the Riasi thrust sheet and the Krol thrust sheet in the east, support the notion that the (Trans-Indus) Salt Range was originally in line with the Himalayan front. It is beyond the resolution of palaeomagnetic data to establish whether the up to 200 km distance between the Salt Range front and the IKSZ, the northwestern continuation of MCT-MBT complex beyond the Western Himalayan Syntaxis, represents the magnitude of southwards movement facilitated by the evaporitic Infra-Cambrian Salt Range Formation or reflects merely a southwards progradation of the detachment front with much smaller displacement.

Secondary components dated at 50 Ma to less than 60 Ma are observed consistently in the pre-Lower Tertiary rocks. These components are comparable to secondary magnetizations widely present in the more internal Himalayan belt and are likewise interpreted to result from thermo-chemical effects during initial collision of Greater India with Asia. This becomes the more acceptable if

one considers the original more internal position of the (Trans-Indus) Salt Range in the PFB-Himalayan orogenic belt. Alternatively, these secondary components may have been acquired during uplift and erosion of an outer rise formed on the underthrusting northern margin of Greater India This outer rise may have prograded southwards to induce at a later stage the 40–50 Ma old secondary components observed in the southern Sulaiman and northern Kirthar Ranges [8]

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